

# Sada Kakarla

Machine Learning Engineer | GenAI, Recommender Systems & Agentic AI

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## Professional Experience

### Handshake

AI Researcher

02/2026 – Present

- Architected a multi-step **agentic coding benchmark**, generating synthetic post-training data across **5+ frontier models** and **30+** custom tasks with rubric-based milestone scoring to drive model improvements for leading AI labs.
- Built automated end-to-end Python evaluation pipelines for **preference ranking** of AI-generated code, replacing manual rubric-based evaluation and powering synthetic alignment data for **SimPO/DPO post-training**.
- Orchestrated scalable **RLVR** post-training simulations to optimize agent policies, combining **veRL (PPO)** and **PyTorch TorchTitan (FSDP)** to train models across interactive **Gymnasium** coding environments and verifying autonomous reasoning improvements via ablation studies.

### Artificial Intelligence Innovation Lab (A2IL)

Applied AI Intern | University at Buffalo | Buffalo, NY

09/2025 – 02/2026

- Engineered a **multimodal AI agent** to automate complex document-extraction workflows, combining **MCP** tool execution and **SigLIP** embeddings for large-scale visual reasoning to reduce hallucinations by **45%** and improve intent classification by **20%**.
- Built a retrieval/ranking system to extract document-image evidence, improving **nDCG@10** by **14%** with **RQ-VAE**, **ColPali**, and **LambdaMART**.
- Trained offline **RL policy (CQL)** and **RLHF** feedback reward model on **5K+** agent trajectories, lifting long-horizon task completion rate by **12%**.

### OnePay

Machine Learning Engineer Intern | New York, NY

06/2025 – 08/2025

- Owned end-to-end development of a **debit rewards recommendation** system across a **50M** transaction marketplace, optimizing for **CTR/CVR** objectives, using **Two-Tower** retrieval and **MMoE** ranking to lift **NDCG@10** by **30%** and reward redemption by **14%** in A/B tests.
- Engineered **40+** behavioral and temporal features in **PySpark** for **CTR/CVR** prediction across users, merchants, and transaction patterns.
- Improved sparse and cold-start user recommendations over **Collaborative Filtering** baselines via **FAISS** and **InfoNCE**, lifting **MRR** by **4%**.
- Identified a user support bottleneck and built an internal **LangGraph AI agent** for intent-aware retrieval, reducing repetitive queries by **27%**.

### Centre for Information Security, Digital Forensics and Cyber Resilience (C-ISFCR)

AI/ML Engineer | Bengaluru, India

05/2022 – 06/2024

- Developed a **GenAI document-intelligence system** on **AWS** for case triage, building a RAG pipeline with **OCR** and **context-aware chunking** across **20K+** heterogeneous records (policy documents, scanned forms, images) to streamline case reviews using citation-backed evidence.
- Designed a **multi-step LLM pipeline** to classify intent and document type, route to specific departments, and retrieve similar historical cases across multimodal inputs, integrating confidence-threshold routing to escalate uncertain queries and reduce triage time by **28%**.
- Architected a **hybrid retrieval system** for precise evidence extraction, combining dense (**BGE-v1.5**) and late-interaction (**ColBERT**) models via Reciprocal Rank Fusion (**RRF**) and cross-encoder re-ranking to achieve a **92% Recall@5**, outperforming a **BM25** baseline.
- Fine-tuned **Llama** via **DPO/QLoRA** on case-resolution pairs, achieving an **88% RAGAS** pass rate across **1K+ evals** for faithfulness & relevance.

### United Nations Academic Impact (MCN)

Software Engineer, AI/ML Systems | Bengaluru, India

10/2021 – 04/2022

- Led the end-to-end fine-tuning and deployment of **Falcon 7B** by architecting scalable **LLM** microservices with FastAPI, **LangChain**, and **LlamaIndex**, increasing automated task completion by **24%** for pediatric follow-up workflows.
- Engineered a **GraphRAG** retrieval pipeline using **Neo4j** and **Cypher** to query complex **medical knowledge graphs** (WHO guidance, care pathways, symptom history), improving factual grounding for Autism Spectrum Disorder (ASD) support queries by **31%**.
- Led evaluation framework design with BMCRI clinicians, aligning on response-validation criteria to cut unsupported AI guidance by **33%**.
- Built an ML triage pipeline (**AWS SageMaker/LightGBM**) with **PHI redaction** and human fallback, improving urgent-case routing by **18%**.

### United Nations Academic Impact (MCN)

Software Engineer, ML Systems | Bengaluru, India

10/2021 – 04/2022

- Built a **GraphRAG** service to identify healthcare supply bottlenecks and capacity risks, orchestrating **Cypher** queries against a **50K-node Neo4j** graph using **LangChain**, FastAPI, and fine-tuned **Llama-2**.
- Automated decision-support pipelines by extracting critical entities from unstructured healthcare logs using **DeBERTa-v3 (NER)**.
- Reduced healthcare surge forecasting RMSE by **18%** over ARIMA baselines using **LightGBM**, live **Kafka** data streams & **Airflow** for retraining.
- Deployed LLMs on **AWS SageMaker** with caching, observability, and **HITL** review to reduce response latency and ensure output safety.

## Technical Skills

**Languages:** Python, C++, TypeScript, SQL, Java, Go, Cypher

**AI & Machine Learning:** PyTorch, TensorFlow, Hugging Face, Transformers, FAISS, Pinecone, scikit-learn, XGBoost, LTR models, LangChain, LangGraph, AutoGen, RAG, ReAct, Tool Calling, RLHF, DSPy, LLM-as-a-Judge, Diffusion Models, Google ADK, LLM APIs (OpenAI, Anthropic)

**Backend & Data:** FastAPI, REST APIs, React, Next.js, Node.js, GraphQL, PySpark, Databricks, Kafka, Elasticsearch, BigQuery, Pydantic, MongoDB

**MLOps & Cloud:** AWS, GCP, Azure, Docker, Kubernetes, vLLM, DeepSpeed, Quantization, MLflow, Airflow, Ray, TensorRT-LLM, Git, CI/CD

**Developer Tools:** Claude Code, GitHub Copilot, Cursor, VS Code, Jupyter, Google Colab, Postman, Terraform

## Projects

**InferRoute – Multi-Provider LLM Inference Gateway | Tech: Go, gRPC, Redis, Kafka, Docker, Kubernetes, AWS EKS, OpenTelemetry**

- Built a **Go**-based inference gateway for long-horizon **agentic workflows**, routing requests across **6 LLM providers** using cached latency, cost, context-window, and tool-use metadata with sub-10ms routing overhead.
- Scaled **5K+** concurrent long-running agent streams with failover, circuit breakers, **Redis** coordination, **Kafka** events & execution observability.

**Monocular ViT 3D Scene Reconstruction | Tech: Python, LangChain, Llama-3, XGBoost, Next.js, TypeScript**

- Built a multi-stage perception pipeline using **ResNet** object detection and **ViT** depth estimation, converting single images into point clouds, meshes, and voxel grids with **85%** mAP on object localization.
- Integrated **Llama-Mesh** for language-driven spatial reasoning over reconstructed 3D scenes, using geometric tool calls for distance, proximity, and occlusion checks to answer **75%** of complex spatial queries correctly.

**GlucoChat – Bilingual Conversational AI Assistant for Diabetic Care | Tech: Python, LangChain, Llama-3, XGBoost, Next.js, TypeScript**

- Designed a **GraphRAG** QA assistant over a **medical knowledge graph** and **NIH biomedical corpora** to answer grounded medical knowledge queries, improving **multi-hop** answer accuracy by **30%** with **BGE-M3** retrieval and **HyDE query rewriting**.
- Reduced unsafe medical recommendations by **32%** with an **XGBoost** safety pipeline, delivered through a **Next.js/TypeScript** interface.

**Temporal Fusion Transformers for Renewable Energy Forecasting | Tech: PyTorch, LSTM, scikit-learn, XGBoost, Open-Meteo API, Streamlit**

- Developed a forecasting system for hourly electricity load across Germany and the UK, integrating **5+** years of **ENTSO-E grid data** with weather signals from the **Open-Meteo API**.
- Trained a **1.4M-parameter Temporal Fusion Transformer** in **PyTorch Lightning** for **24-hour** multi-horizon load forecasting, achieving **3.1% validation SMAPE** with **LSTM/ARIMA** baselines, interpretable attention, variable selection, and a **Streamlit** dashboard for visualization.

**NegotiableAI – Hierarchical RL Agent for Autonomous Deal Negotiation | Tech: Python, LangGraph, PyTorch, Ray, PPO, Double DQN**

- Developed a hierarchical RL negotiation agent via **LangGraph**, pairing a **Double DQN** intent-selector with a **Qwen-2.5 7B** dialogue model trained via **PPO** self-play and **RLHF reward modeling**, boosting agreement rates from **40%** to **60%**.
- Engineered Pydantic guardrails across **Ray**-distributed simulations to enforce pricing/discount rules, lifting constraint compliance by **18%**.

**Multi-Modal VQA System for Early Autism Spectrum Disorder Analysis | Tech: Python, PyTorch, OpenCV, LSTM, BERT, CNNs**

- Architected a **Visual Question Answering** system to analyze child interaction videos for early ASD indicators by combining facial expression recognition, gaze patterns, visual behavioral cues, and textual question-response signals.
- Engineered a temporal deep learning pipeline using a **dual LSTM** to model long-term behavioral dependencies across **video sequences**, achieving **98%** predictive accuracy and earned **Best Capstone Project** recognition for its clinical and social impact.

**Edge Vision System for Drone & Traffic Anomaly Detection | Tech: YOLO, OpenCV, ByteTrack, FastAPI, Docker, ONNX/TensorRT**

- Built a real-time edge computer vision pipeline to detect, track, and classify vehicles, pedestrians, drones, and anomalous motion patterns from live or recorded video streams using **YOLO**, **OpenCV**, and **ByteTrack/DeepSORT**.
- Hardened edge deployment with robustness testing for low light, motion blur, dropped frames, and occlusion, then served the model through **FastAPI** and **ONNX/TensorRT** with latency monitoring and model versioning.

## Education

**Master of Science in Artificial Intelligence**

University at Buffalo - SUNY Buffalo

08/2024 – 12/2025

Buffalo, NY

**Bachelor of Technology in Computer Science and Engineering**

PES University

12/2020 – 05/2024

Bengaluru, India